

Simulation Purpose Definition Module (SPDM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Purpose & Scope Standard (SPSS)

Operational Layer: Simulation Purpose & Scope Governance Layer

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Governed Space: Simulation Purpose Definition

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Canonical Language: English (UCL™)

Governed Space

Simulation Purpose Definition

Canonical Definition

Simulation Purpose Definition Module (SPDM) defines the governance conditions under which the reason for performing a simulation is explicitly identified, justified, bounded, documented, and connected to the Simulation Object and the question the simulation is intended to investigate.

SPDM establishes the purpose basis from which simulation scope, applicability, required depth, precision, reality representation, assumptions, scenarios, and downstream simulation design can legitimately be derived.

Operational Role

SPDM serves as the first internal governance space of the Simulation Purpose & Scope Standard (SPSS).

The preceding Simulation Object Standard establishes:

What exactly is being simulated?

SPDM asks:

Why is this Simulation Object being simulated, and what legitimate inquiry is

the simulation intended to support?

The governed progression begins:

Simulation Object → Simulation Need → Simulation Purpose → Simulation Question
→ Intended Outcome → Purpose Boundary

A simulation should not proceed into detailed representation or design before this purpose basis is sufficiently established.

Module Operational Space

SPDM governs: Simulation Purpose identification, Simulation Need, purpose justification, Simulation Question formation, intended simulation outcome, intended analytical use, intended decision-support role, exploratory versus confirmatory purpose, design-support purpose, testing purpose, comparison purpose, safety-related purpose, training purpose, predictive or scenario-exploration purpose, pre-deployment simulation purpose, purpose hierarchy, multiple-purpose simulations, purpose compatibility, purpose conflicts, purpose changes, purpose expansion, purpose substitution, purpose traceability, and Purpose Definition Records.

Module Function

The function of SPDM is to ensure that simulation begins from an explicit governed purpose rather than from available technology, an existing model, available data, organizational habit, or an undefined desire to “simulate the system.”

SPDM prevents: simulations without a defined reason, technical capability determining purpose retrospectively, vague simulation objectives, hidden secondary purposes, conflicting purposes being treated as one inquiry, exploratory simulations being represented as confirmatory, simulation results being used for decisions the simulation was never designed to support, purpose expansion after results are known, and downstream simulation architecture being constructed before the intended inquiry is understood.

Simulation Need

Before defining purpose, an organization should establish why simulation is an appropriate activity.

A Simulation Need exists where examination of a Simulation Object through governed simulated representation can materially contribute to understanding, testing, comparison, exploration, preparation, design, or decision support.

The existence of a Simulation Object does not itself establish a need to simulate it.

Simulation Purpose

Simulation Purpose defines the primary reason the simulation is being performed.

Examples may include:

Exploration — understanding possible behaviors or outcomes.

Testing — examining behavior against defined conditions.

Comparison — comparing configurations, strategies, systems, or alternatives.

Design Support — informing development or engineering decisions.

Safety Investigation — examining behavior under potentially hazardous conditions.

Operational Preparation — studying expected operational situations before deployment.

Training — reproducing governed situations for human or system preparation.

Scenario Analysis — examining alternative future or hypothetical conditions.

Decision Support — producing simulation-derived information relevant to a defined decision.

These purposes may coexist, but they must remain distinguishable where their governance requirements differ.

Purpose Must Precede Simulation Design

SIMULOS® establishes:

The simulation purpose should determine the simulation design; the available simulation design should not silently determine the purpose.

An organization may already possess a model, digital twin, simulator, dataset, environment, or simulation platform.

Their availability does not establish that they are appropriate for the new purpose.

Simulation Question

Every material Simulation Purpose should be translated into one or more explicit Simulation Questions.

For example:

Purpose:

Evaluate autonomous robot behavior before deployment.

Simulation Question:

How does the defined robot configuration behave under the specified operating conditions and disturbance scenarios?

The Simulation Question creates a bridge between general purpose and governable scope.

Question Specificity

A Simulation Question should be sufficiently specific to determine what downstream simulation design must investigate.

A question such as:

“Is the system good?”

is not sufficiently governed.

A question such as:

“Can the specified robotic configuration complete the defined material-handling task within the modeled operating envelope while remaining within the specified operational constraints?”

creates a substantially stronger simulation basis.

Intended Outcome

SPDM should identify what type of outcome the simulation is expected to produce.

The intended outcome may include: behavioral understanding, comparative information, performance observations, scenario outcomes, design information, failure observations, boundary observations, sensitivity information, operational preparation, or decision-support information.

The intended outcome should not predetermine the substantive result. Purpose ≠ Desired Result

A simulation may have a legitimate intended purpose.

It must not have a predetermined required conclusion.

For example:

Purpose: Determine whether Configuration A performs better than Configuration B

is legitimate.

Purpose: Demonstrate that Configuration A is better than Configuration B

may create methodological bias if the simulation is structurally expected to confirm that conclusion.

SPDM therefore separates:

What the simulation should investigate

from:

What the organization wants the simulation to conclude.

Exploratory Simulation

An exploratory simulation investigates possibilities without necessarily testing a predetermined proposition.

It may be appropriate for: discovery, sensitivity exploration, scenario generation, early design, emerging-risk investigation, or behavior mapping.

Exploratory results should remain identified as exploratory when downstream reliance is considered.

Confirmatory Simulation

A confirmatory simulation evaluates a defined proposition, requirement, expectation, configuration, or hypothesis against governed conditions.

Its purpose should identify what is being examined and against which intended question.

Confirmatory simulation normally requires stronger control over: scope, assumptions, scenarios, representation, inputs, execution, and interpretation.

Decision-Support Simulation

Where simulation is intended to support a decision, SPDM should identify the decision context.

The record should distinguish between:

Simulation informs the decision

and:

Simulation determines the decision.

These are materially different governance conditions.

High-Consequence Purpose

Where simulation may materially influence: human safety, critical infrastructure, autonomous deployment, healthcare, financial exposure, regulatory decisions, public systems, or other high-consequence environments,

the purpose should explicitly identify that consequence context.

This information may affect downstream Simulation Depth and governance requirements.

Pre-Deployment Purpose

SIMULOS® may be used to examine systems before real-world deployment.

For pre-deployment simulation, the purpose should identify what the

organization seeks to understand before exposure to operational reality.

Examples include: behavioral boundaries, failure conditions, environmental response, interaction behavior, autonomous decisions, performance limitations, or scenario response.

SPDM does not determine whether simulation alone is sufficient for deployment authorization.

Multiple Simulation Purposes

One simulation may support multiple purposes.

For example:

Design Optimization + Safety Exploration + Operator Training

These purposes should not automatically be merged into one undifferentiated purpose.

Each may require different: scope, representation, assumptions, scenarios, precision, outputs, and downstream reliance boundaries.

Purpose Hierarchy

Where several purposes exist, SPDM may establish:

Primary Purpose

Secondary Purpose

Supporting Purpose

This helps prevent a secondary use from silently becoming the primary justification for the simulation.

Purpose Compatibility

Multiple purposes may share one simulation only where their requirements are sufficiently compatible.

If one purpose requires substantially different representation, precision, scenarios, or operating conditions, separate simulation pathways may be necessary.

Purpose Conflict

A Purpose Conflict exists where two or more simulation purposes create incompatible methodological requirements.

For example:

A simulation optimized for rapid exploratory iteration may not simultaneously

satisfy the conditions required for a high-consequence confirmatory simulation.

Purpose conflicts should be resolved or explicitly separated before downstream simulation design.

Purpose Boundary

Every Simulation Purpose should establish a boundary around what the simulation is intended to investigate.

The purpose boundary prevents the simulation from later being interpreted as having addressed questions that were never part of the original inquiry.

Conceptually:

Legitimate Simulation Interpretation $\leq$ Governed Simulation Purpose Purpose Expansion

A Simulation Purpose may legitimately expand during a project.

However:

Purpose expansion requires governance reassessment.

A broader purpose may require changes to: scope, representation, assumptions, inputs, scenarios, depth, precision, execution, or downstream interpretation.

Existing simulation work should not automatically be presumed sufficient for the expanded purpose.

Purpose Reduction

Purpose may also narrow.

Where this occurs, downstream simulation activities may be simplified only where the reduced purpose no longer requires them.

Historical purpose should remain traceable.

Purpose Substitution

Purpose Substitution occurs when simulation results generated for one purpose are later represented as supporting a materially different purpose.

Example:

Original Purpose: Early Engineering Exploration

later becomes:

Claimed Purpose: Deployment Safety Demonstration

without corresponding simulation redesign or reassessment.

SIMULOS® treats this as a material governance issue.

Purpose Change

A material change in purpose should create a Purpose Reassessment Trigger.

The reassessment should determine whether existing: scope, assumptions, representations, scenarios, data, execution, and outputs

remain suitable.

Purpose Traceability

SPDM should preserve the relationship:

Simulation Object → Simulation Need → Simulation Purpose → Simulation Question
→ Intended Outcome

This relationship should remain traceable throughout the simulation lifecycle.

Downstream simulation outputs should therefore be interpretable against the purpose that originally justified their production.

Purpose and Reality Representation

SPDM does not determine how reality should be represented.

It establishes why reality must be represented and what inquiry that representation must support.

The downstream Reality Representation governance space determines how the relevant aspects of reality are represented for that purpose.

Purpose and Validation

SPDM does not validate the simulation.

VALIDOS® governs general validation methodology where validation is required.

SIMULOS® establishes the simulation-specific purpose that may later become part of a validation inquiry.

This preserves the boundary:

SIMULOS® governs the simulation.

VALIDOS® governs validation of the relevant object or claim.

Purpose and Integrity

SPDM does not govern general integrity.

INTEGROS® governs integrity where applicable.

SPDM governs whether the simulation purpose itself remains explicit, bounded, stable, and traceable.

Failure Conditions

SPDM should identify a governance failure where: no explicit Simulation Purpose exists, purpose cannot be connected to the Simulation Object, the Simulation Question remains materially ambiguous, the intended outcome is undefined, purpose is defined retrospectively from available results, a desired result is disguised as simulation purpose, materially different purposes are silently merged, incompatible purposes are not resolved, purpose expands without reassessment, purpose substitution occurs, high-consequence use is not reflected in the purpose, or downstream simulation use materially exceeds the governed purpose.

Minimum Implementation Framework

1. Identify the Simulation Need

Document why simulation is being used for the defined Simulation Object. 2. Define the Simulation Purpose

State the primary reason for performing the simulation and distinguish secondary purposes where applicable. 3. Establish the Simulation Question

Translate the purpose into a sufficiently explicit question or set of questions that downstream simulation design can address. 4. Define the Intended Outcome and Purpose Boundary

Identify what type of information the simulation is expected to produce and what the simulation is not intended to establish. 5. Approve and Preserve Purpose Traceability

Record the governed purpose basis and maintain its connection to downstream scope, representation, design, execution, outputs, and any subsequent purpose changes.

Governance Outputs

SPDM may produce: Simulation Need Record, Simulation Purpose Statement, Primary Simulation Purpose, Secondary Simulation Purpose Record, Simulation Question Set, Intended Simulation Outcome, Decision-Support Context Record, Consequence Context Record, Purpose Hierarchy, Purpose Compatibility Assessment, Purpose Conflict Record, Purpose Boundary, Purpose Change Record, Purpose Expansion Record, Purpose Substitution Record, Purpose Reassessment Trigger, and Simulation Purpose Traceability Record.

Use Case 1 — Autonomous Robot Pre-Deployment Simulation

Scenario

An industrial organization intends to deploy an autonomous mobile robot within a production facility.

The organization creates a simulation before physical deployment. Application

SPDM establishes the primary purpose as:

Pre-deployment investigation of autonomous robot behavior within defined production operating conditions.

The governed Simulation Questions include whether the robot can: navigate the intended environment, respond to defined obstacles, operate within specified boundaries, and respond to selected abnormal conditions.

SPDM explicitly prevents the simulation from being represented as universal proof of safety across environments not included in the purpose. Result

The downstream simulation architecture receives a defined inquiry basis from which scope, reality representation, assumptions, scenarios, depth, and precision can be established. Use Case 2 — Aerospace Design Comparison

Scenario

An aerospace engineering team uses simulation to compare two proposed component configurations before physical prototyping. Application

SPDM defines:

Primary Purpose: Comparative engineering evaluation.

Simulation Question: How do Configurations A and B differ under the defined modeled operating conditions?

Intended Outcome: Comparative performance and behavior information supporting design selection.

The purpose does not state that Configuration A must outperform Configuration B. Result

Both configurations can be simulated against a common governed purpose without predetermining the result, while later use of the simulation for unrelated certification or deployment claims would require separate governance assessment.

Architectural Position

Simulation Purpose Definition is the first internal governance space of the Simulation Purpose & Scope Standard (SPSS).

It follows the establishment of the Simulation Object and precedes detailed governance of Simulation Scope and downstream Reality Representation.

The architectural progression is:

Simulation Object → Simulation Need → Simulation Purpose → Simulation Question → Intended Outcome → Purpose Boundary → Simulation Scope → Reality Representation

SPDM therefore establishes the reason from which the remainder of the

simulation governance lifecycle begins.

Foundational Principle

A simulation can legitimately support only the inquiry for which its purpose was explicitly defined and governed.

Canonical Closing Statement

Simulation Purpose Definition Module establishes the governed purpose foundation of SIMULOS® by defining why a Simulation Object is being simulated, which questions the simulation is intended to investigate, what type of outcome it is expected to produce, and where the legitimate purpose boundary ends. Through governance of simulation need, purpose definition, question formation, intended outcomes, multiple-purpose compatibility, purpose conflict, purpose change, expansion, substitution, consequence context, and complete purpose traceability, SPDM ensures that simulation design begins from a defined inquiry and that simulation results cannot legitimately acquire a broader purpose merely because downstream users wish to apply them beyond the reason for which the simulation was originally created.

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